



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
FULL SYLLABUS
09-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

SPACE FOR ROUGH WORK

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

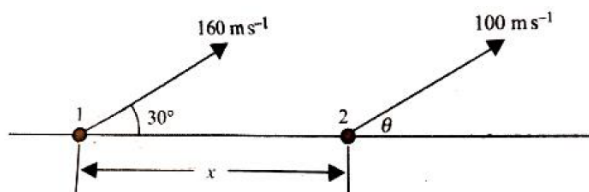
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

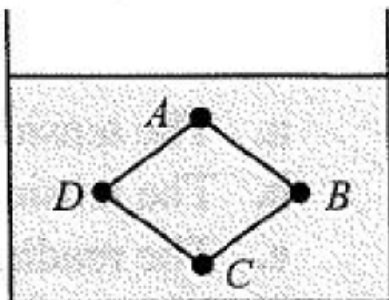
1. Suppose two particles 1 and 2 are projected in vertical plane simultaneously.



Their angles of projection are 30° and θ , respectively, with the horizontal. Let them collide after a time t in air. Then (neglect air resistance)

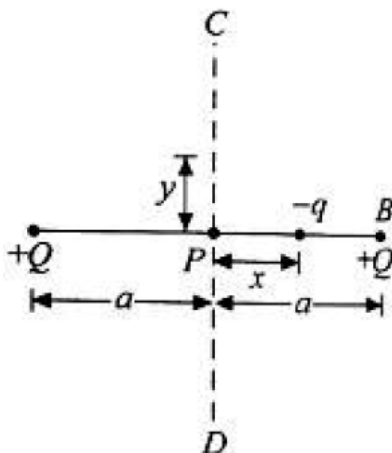
- (A) $\theta = \sin^{-1}(4/5)$ and they will have same speed just before the collision.
- (B) $\theta = \sin^{-1}(4/5)$ and they will have different speed just before the collision.
- (C) $x < 1280\sqrt{3} - 960\text{m}$
- (D) It is possible that the particles collide when both of them are at their highest point.

2. Figure shows a container filled with a liquid of density ρ . Four points A, B, C and D lie on the vertices of a vertical square. Points A and C lie on a vertical line and points B and D lie on a horizontal line. Choose the correct statement(s) about the pressure at the four points.



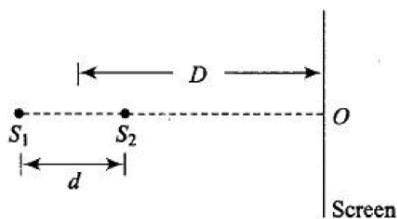
- (A) $P_D = P_B$
 (B) $P_A < P_B = P_D < P_C$
 (C) $P_D = P_B = \frac{P_C - P_A}{2}$
 (D) $P_D = P_B = \frac{P_C + P_A}{2}$

3. In the arrangement shown in figure, two positive charges, $+Q$ each, are fixed. Mark the correct statement(s) regarding a third charged particle $-q$ placed at the midpoint P that can be displaced along or perpendicular to the line connecting the charges.

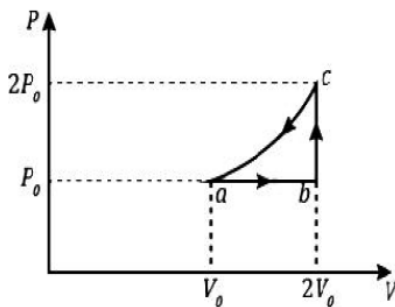


- (A) The particle will perform SHM for $x \ll a$.
- (B) The particle will oscillate about P but not harmonically for any x .
- (C) The particle will perform SHM for $y \ll a$.
- (D) The particle will oscillate about P but not harmonically for y comparable to a .
4. A charged particle P leaves the origin with speed $v = v_0$ at some inclination with the x -axis. There is a uniform magnetic field B along the x -axis. P strikes a fixed target T on the x -axis for a minimum value of $B = B_0$. P will also strike T if.
- (A) $B = 2B_0, v = 2v_0$
- (B) $B = 2B_0, v = v_0$
- (C) $B = B_0, v = 2v_0$
- (D) $B = \frac{B_0}{2}, v = 2v_0$

5. Two points monochromatic and coherent sources of light of wavelength λ each are placed as shown in figure. The initial phase difference between the sources is zero. ($D \gg d$). Mark the correct statement(s).



- (A) If $d = \frac{7\lambda}{2}$, O will be a minima.
- (B) If $d = \lambda$, only one maxima can be observed on the screen.
- (C) If $d = 4.8\lambda$, then total 5 minima would be there on the screen.
- (D) If $d = \frac{5\lambda}{2}$, the intensity at O would be minimum.
6. One mole of an ideal monatomic gas has initial temperature T_0 , is made to go through the cycle abca shown in the given figure. If U denotes the internal energy, then choose the correct alternative.



- (A) $U_c > U_b > U_a$
- (B) $U_c - U_b = 3RT_0$
- (C) $U_c - U_a = \frac{9RT_0}{2}$
- (D) $U_b - U_a = \frac{3RT_0}{2}$

SECTION-I (ii) : (Maximum Marks: 12)

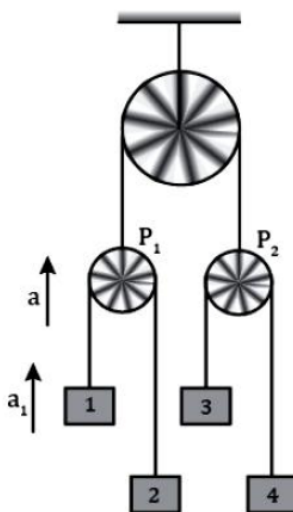
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. In the system shown in figure, masses of the blocks are such that when system is released, the acceleration of pulley P_1 is a upwards and the acceleration of block 1 is a_1 upward. It is found that the acceleration of block 3 is same as that of 1 both in magnitude and direction. Given that $a_1 > a > a_1/2$. Match the following.



List-I		List-II	
(P)	Acceleration of 2	(1)	$2a + a_1$
(Q)	Acceleration of 4	(2)	$2a - a_1$
(R)	Acceleration 2 w.r.t. 3	(3)	Upward
(S)	Acceleration 2 w.r.t. 4	(4)	Downward

(A) $P \rightarrow 2,3; Q \rightarrow 1,4; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 1,4; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 2,3; Q \rightarrow 1,2; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

8. Bodies A and B are rolling without slipping on a rough inclined plane, starting from rest.

Match the following.

List -I		List -II	
(P)	Velocity of A and B after travelling the same distance will be	(1)	Different if A and B have same size but different shape and mass
(Q)	Time taken by A and B to reach the bottom of the plane from the same height will be	(2)	Different if A and B have same size and same mass but having different shapes
(R)	Acceleration of centre of mass of A and B will be	(3)	Same if A and B have same shape and same size but of different materials
(S)	During rolling motion, frictional force experienced by A and B will be	(4)	Same if A and B have same size but different shape and mass

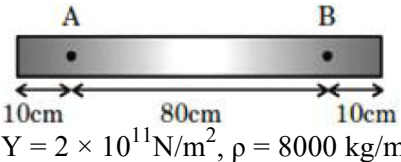
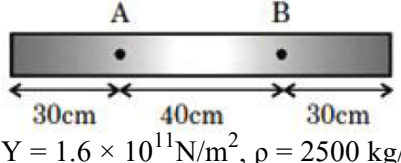
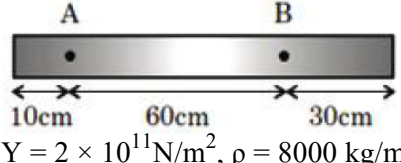
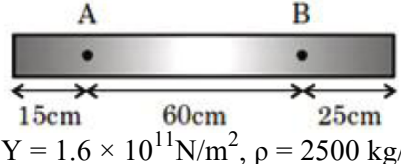
(A) $P \rightarrow 1,2,3; Q \rightarrow 2,3,4; R \rightarrow 1,2,3; S \rightarrow 1,2$

(B) $P \rightarrow 1,2,3; Q \rightarrow 1,2,3; R \rightarrow 1,2; S \rightarrow 1,2,4$

(C) $P \rightarrow 1,2,3; Q \rightarrow 2,3,4; R \rightarrow 1,2; S \rightarrow 1,2$

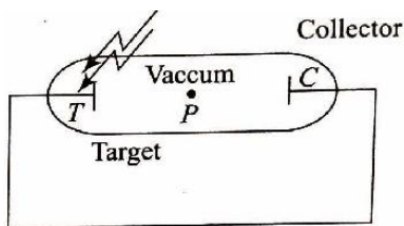
(D) $P \rightarrow 1,2,3; Q \rightarrow 1,2,3; R \rightarrow 1,2,3; S \rightarrow 1,2$

9. List-I represents the longitudinal vibration of metallic rods each of length 1m and each rod is rigidly clamped at two points A and B. List-II represents their frequency and wavelength of longitudinal vibration in the fundamental mode of vibration. Then match the option of List-I correctly with the options of List-II.

List -I		List -II	
(P)	 $Y = 2 \times 10^{11} \text{ N/m}^2, \rho = 8000 \text{ kg/m}^3$	(1)	$f = 40 \text{ kHz}$
(Q)	 $Y = 1.6 \times 10^{11} \text{ N/m}^2, \rho = 2500 \text{ kg/m}^3$	(2)	$f = 12.5 \text{ kHz}$
(R)	 $Y = 2 \times 10^{11} \text{ N/m}^2, \rho = 8000 \text{ kg/m}^3$	(3)	$f = 20 \text{ kHz}$
(S)	 $Y = 1.6 \times 10^{11} \text{ N/m}^2, \rho = 2500 \text{ kg/m}^3$	(4)	$\lambda = 0.4 \text{ m}$
		(5)	$\lambda = 0.2 \text{ m}$

- (A) $P \rightarrow 1,5; Q \rightarrow 3,4; R \rightarrow 2,5; S \rightarrow 1,5$
- (B) $P \rightarrow 2,5; Q \rightarrow 3,5; R \rightarrow 2,4; S \rightarrow 2,5$
- (C) $P \rightarrow 3,4; Q \rightarrow 3,5; R \rightarrow 2,4; S \rightarrow 2,5$
- (D) $P \rightarrow 2,4; Q \rightarrow 3,4; R \rightarrow 2,4; S \rightarrow 1,5$

10. In a photoelectric experimental arrangement, light of frequency f is incident on a metal target whose work function is $\phi = \frac{hf}{3}$ as shown. In list-I, KE of photoelectron is mentioned at various locations/instants and in list-II, the corresponding values. Match the entries of list-I with the entries of list-II.



List -I		List -II	
(P)	Maximum KE of photoelectrons just after emission from target	(1)	Zero
(Q)	KE of photoelectrons just after emission from target	(2)	$hf/3$
(R)	KE of photoelectron when they are halfway between the target and collector	(3)	$hf/2$
(S)	KE of photoelectrons as they reach the collector	(4)	$2hf/3$

- (A) $P \rightarrow 4; Q \rightarrow 1,2,3,4; R \rightarrow 2,3,4; S \rightarrow 1$
- (B) $P \rightarrow 4; Q \rightarrow 1,4; R \rightarrow 2,3; S \rightarrow 1,3,4$
- (C) $P \rightarrow 1,4; Q \rightarrow 4; R \rightarrow 3,4; S \rightarrow 1,3,4$
- (D) $P \rightarrow 4; Q \rightarrow 1,2,3,4; R \rightarrow 1,2,3,4; S \rightarrow 1,2,3,4$

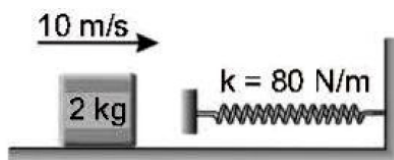
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

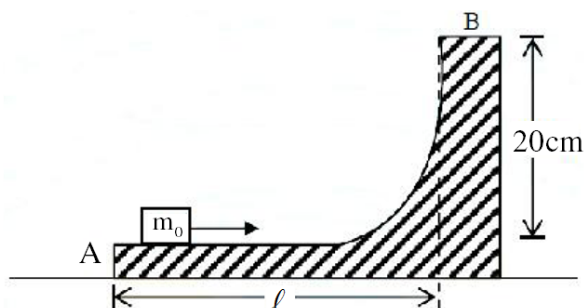
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

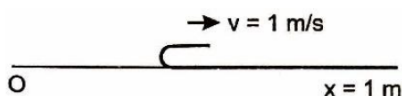
1. A car travelling at 28 m/s has no tendency to slip (i.e. assume friction force on the car = 0) on the track of radius 200m banked at an angle θ . When the speed is increased to 35 m/s, the car is just on the point of slipping up the track. Calculate the coefficient of friction between the car and the track. (Upto two decimals) ($g = 10 \text{ m/s}^2$.)
2. A block of mass 2kg approaches a spring of stiffness $k = 80 \text{ N/m}$ on a smooth horizontal plane with a speed of 10 m/s. Find the time in seconds (upto 2 decimals) in which kinetic energy of the block reduces to 50% of its initial kinetic energy. Assume $t = 0$ when the block just touches the spring, for the first time.



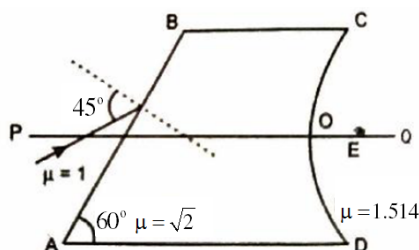
3. A wedge of mass $M = 2m_0$ rests on a smooth horizontal plane. A small block of mass m_0 rests over it at left end A as shown in figure. A sharp impulse is applied on the block, due to which it starts moving to the right with velocity $v_0 = 6 \text{ m/s}$. At highest point of its trajectory, the block collides with a particle of same mass m_0 moving vertically downwards with velocity $v = 2 \text{ m/s}$ and gets stuck with it. If the combined mass lands at the end point B of the body of mass M, calculate length ℓ . Neglect friction, take $g = 10 \text{ m/s}^2$.



4. A long thin carpet is laid on the floor. One end of the carpet is bent back and then pulled backwards with constant velocity $v = 1 \text{ m/s}$, just above the part of the carpet which is still at rest on the floor. What is the minimum force (in SI system) needed to pull the moving part, if the carpet has unit length and unit mass. (Neglect all dissipative forces) (upto 1 decimal)



5. The volume density of the elastic deformation energy in fresh water at a depth of $h = 1 \text{ km}$. (Take density of water (constant) $= 10^3 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$ and bulk modulus of elasticity of water $B = 2 \times 10^9 \text{ N/m}^2$) is $x \times 10^4 \text{ J/m}^3$. Find 'x' (in 2 digit) (neglect atmospheric pressure).
6. Figure shows an irregular block of material of refractive index $\sqrt{2}$. A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4 m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE in meters (upto two places of decimal)



7. A photon with an energy of 4.9 eV ejects photoelectrons from tungsten. When the ejected electron enters a constant magnetic field of strength $B = 2.5 \text{ mT}$ at an angle of 60° with the field direction, the maximum pitch of the helix described by the electron is found to be 2.7 mm . Find the work function of the metal in electron volts (upto two digits). Given that specific charge of electron is $1.76 \times 10^{11} \text{ C/kg}$
8. Consider the β^- decay $^{198}\text{Au} \rightarrow ^{198}\text{Hg} + \beta^- + \bar{\nu}$
Where ^{198}Hg represents a mercury nucleus in an excited state of energy 1.088 MeV above the ground state. What can be the maximum kinetic energy (MeV) of electron emitted (upto two decimals). The atomic mass of ^{198}Au is 197.968233 u and that of ^{198}Hg is 197.966760 u . $1 \text{ u} = 931 \text{ MeV}/c^2$

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

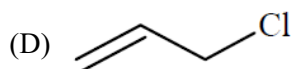
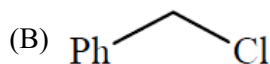
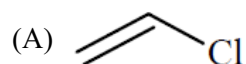
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

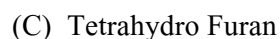
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Which can form square planar complex with coordination number (CN) is 4?
(A) Pt(II), Pd(II) with strong and weak field ligand
(B) Au(III) with strong and weak field ligand
(C) Ni(II) with C.N. = 4 and strong field ligand
(D) Hg(II) with C.N. = 4 and strong field ligand
 2. Froth floatation:
(A) is physical method of separation of mineral from the gangue
(B) is a method to concentrate the ore depending on the difference in wettability of gangue and the ore
(C) is used for the sulphide ores
(D) is a method in which impurities sink to the bottom

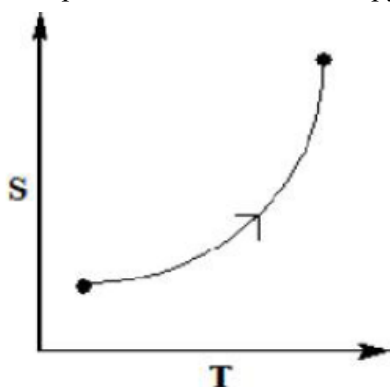
3. Both S_N1 and S_N2 reactions are not favourable in which of the given compounds?



4. Which of the following molecule(s) have higher resonance energy than 1,3-butadiene?



5. A mixture of fixed number of moles of ideal gases undergoes a thermodynamic process as shown by the plot. Which of the following need not be same for the process, if the composition of ideal gas mixture changes for the same initial and final temperatures? S = molar entropy of the mixture, T = temperature



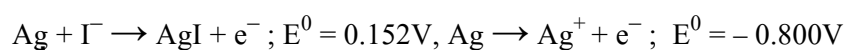
(A) Change in internal energy

(B) Change in free energy

(C) Work done

(D) Change in enthalpy

6. Given the data at 25°C



What is the value of $\log K_{\text{sp}}$ for AgI? $\left(2.303 \frac{RT}{F} = 0.059\text{V}\right)$

(A) - 8.12

(B) + 8.612

(C) - 37.83

(D) - 16.13

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the property of the compound given in List-II with the compound given in List-I.

List – I		List– II	
(A)	KHCO_3	(P)	contain polymeric anion
(B)	NaHCO_3	(Q)	contain dimeric anion
(C)	LiHCO_3	(R)	contain hydrogen bond
(D)	$\text{Mg}(\text{HCO}_3)_2$	(S)	liberates CO_2 on heating
		(T)	can be prepared in solid state

- (A) $A \rightarrow \text{QRST}; B \rightarrow \text{PRST}; C \rightarrow \text{S}; D \rightarrow \text{ST}$
- (B) $A \rightarrow \text{QRST}; B \rightarrow \text{PST}; C \rightarrow \text{PS}; D \rightarrow \text{RST}$
- (C) $A \rightarrow \text{QST}; B \rightarrow \text{PST}; C \rightarrow \text{S}; D \rightarrow \text{RST}$
- (D) $A \rightarrow \text{QST}; B \rightarrow \text{PRST}; C \rightarrow \text{PS}; D \rightarrow \text{ST}$

8. Match the compound given in List-I with the character given in List-II.

List-I		List-II	
(A)	Hydrogen hypo phosphate ion $\text{H}_2\text{P}_2\text{O}_6^{2-}$	(P)	All hydrogens are ionizable
(B)	Hydrogen pyrophosphate ion $\text{H}_2\text{P}_2\text{O}_7^{2-}$	(Q)	Number of acidic hydrogens present is 2
(C)	Pyrosulphurous acid $\text{H}_2\text{S}_2\text{O}_5$	(R)	sp^3 hybridised central atom
(D)	Hydrogen peroxy mono sulphate ion HSO_5^-	(S)	Overall non-planar structure
		(T)	Can act as oxidizing agent

(A) $A \rightarrow \text{PQRS}; B \rightarrow \text{RST}; C \rightarrow \text{PQRS}; D \rightarrow \text{PQRST}$

(B) $A \rightarrow \text{PQS}; B \rightarrow \text{RST}; C \rightarrow \text{PRS}; D \rightarrow \text{PRST}$

(C) $A \rightarrow \text{PQS}; B \rightarrow \text{RS}; C \rightarrow \text{PRS}; D \rightarrow \text{PQRST}$

(D) $A \rightarrow \text{PQRS}; B \rightarrow \text{PQRS}; C \rightarrow \text{PQRS}; D \rightarrow \text{PRST}$

9. Match the correct reduction method from List-II for the reaction in List-I.

List-I		List-II	
(A)		(P)	Birch reduction
(B)		(Q)	Stephen's reduction
(C)		(R)	Wolff-Kishner reduction
(D)		(S)	Clemmensen reduction

(A) $A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow RS$

(B) $A \rightarrow R; B \rightarrow P; C \rightarrow RS; D \rightarrow S$

(C) $A \rightarrow P; B \rightarrow RS; C \rightarrow S; D \rightarrow R$

(D) $A \rightarrow RS; B \rightarrow S; C \rightarrow R; D \rightarrow P$

10. Match the crystal lattice in List-I with the corresponding Geometrical or Physical property in List-II.

List - I (Crystal lattice)		List - II (Geometrical or Physical property)	
(A)	Rock salt unit cell	(P)	Three-fold axis of symmetry
(B)	Zinc blende unit cell	(Q)	Four-fold axis of symmetry
(C)	Fluorite unit cell	(R)	Plane of symmetry
(D)	FCC unit cell	(S)	Anisotropy
		(T)	Number of first nearest identical species is 12

(A) $A \rightarrow PQST; B \rightarrow PQRS; C \rightarrow PRST; D \rightarrow PQRST$

(B) $A \rightarrow PQRST; B \rightarrow PQRST; C \rightarrow PQRS; D \rightarrow PQRST$

(C) $A \rightarrow PQRST; B \rightarrow QRS; C \rightarrow PQRST; D \rightarrow PQRST$

(D) $A \rightarrow PQRST; B \rightarrow PQRST; C \rightarrow PQRT; D \rightarrow PQRT$

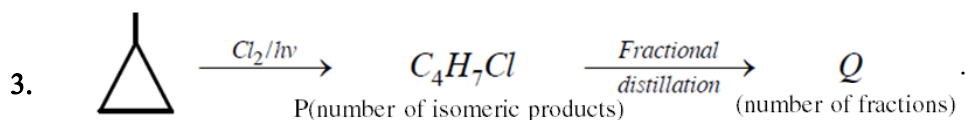
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

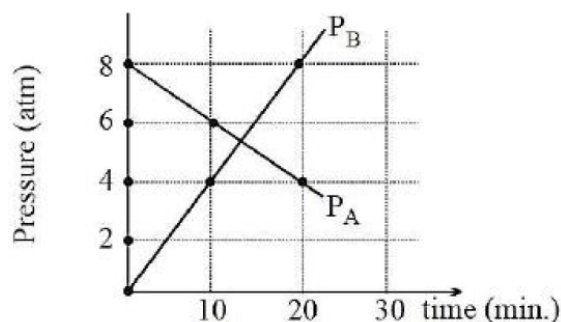
1. A mixture contains one gram equivalent of each of the following metal hydroxides. Water is added to the mixture, boiled and filtered. How many of the hydroxides remain in the residue?
NaOH, KOH, Ba(OH)₂, TiOH, Be(OH)₂, Mg(OH)₂, Al(OH)₃, Sn(OH)₂, Zn(OH)₂, Sr(OH)₂.
2. Polyphosphates are the condensed products of tetrahedral phosphate units that contain P – O – P bonds. A sample of poly phosphoric acid was dissolved in water and titrated with dilute NaOH (aq). Two stoichiometric points were observed at 30.4 and 45.6 cm³. The number of P – O – P bonds in that poly phosphoric acid is



Find the value of P + Q .

4. An organic compound (X) shows following characteristics
 - i) It is optically active
 - ii) Gives +ve haloform test
 - iii) Gives violet colour with neutral FeCl₃ solution
 Then the minimum possible molecular weight of compound (X) is

5. How many compounds are more reactive than ethene towards electrophilic addition reaction?
- (I) $\text{CH}_3 - \text{CH} = \text{CH}_2$
- (II) $\begin{array}{c} \text{O} \\ || \\ \text{H} - \text{C} - \text{CH} = \text{CH}_2 \end{array}$
- (III) $\text{Cl} - \text{CH} = \text{CH}_2$
- (IV) $\text{CH}_3 - \text{O} - \text{CH} = \text{CH}_2$
- (V) $\text{H} - \text{C} \equiv \text{C} - \text{H}$
- (VI) $\text{Ph} - \text{CH} = \text{CH}_2$
- (VII) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$
6. 2 mole NO and 16 gm O_2 were taken in a 6.25 L vessel at 27°C . Calculate the change in pressure when half of the maximum amount of NO_2 has been produced according to reaction.
- (Take $R = \frac{1}{12}$ litre.atm/mol K)
- $$2\text{NO}_{(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{2(\text{g})}$$
7. The energy of the first orbit of hydrogen atom is -13.6 eV. If the fourth ionization enthalpy of gaseous isolated beryllium atom is expressed as $Y \times 10^{-19}$ J, what is the value of 'Y'? Assume nucleus is at rest in either cases. ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)
8. In a chemical reaction where $\text{A}(\text{g})$ is converting to $\text{B}(\text{g})$, the following observation is made. What is the time of completion for the reaction (in min)?



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ is a differentiable function such that $\int_1^{xy} f(t)dt = y \int_1^x f(t)dt + x \int_1^y f(t)dt \forall x, y \in \mathbb{R} - \{0\}, f(1) = 1$.

Which of the following is correct?

(A) $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = 0$

(B) $\int_{1/e}^1 f(x)dx = \frac{1}{e}$

(C) Number of solutions of $f(x) = f^{-1}(x)$ for $x > 0$ is 1

(D) Number of functions satisfying the given conditions can not be finite

2. For a given hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$. Let P is any point on it such that if a tangent drawn at P is made to meet its conjugate axis in 'T'. PN is the ordinate of point P and PT meets transverse axis in 'T¹', then (A, A¹ are the vertices with positive and negative x-co ordinates and C is its centre).

(A) CT.PN = 4

(B) CN.CT¹ = 9

(C) CN.NT¹ = AN.A¹N

(D) CN.NT¹ = CN² + AC²

3. Let a, b, c be integers and $a^2 + b^2 + c^2 = 2$. Let $A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ consider the system of equations

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Then which of the following statement(s) is/are correct -

- (A) The probability that the system of equations has unique solution is 1/2
 (B) The probability that the system of equations has infinitely many solutions is 1/2
 (C) If a = 0, then the number of ordered pairs (b, c) for which the system has no solution is 2
 (D) The number of integers in the range of ab + bc + ca is 2

4. The plane containing the line $\frac{x-3}{2} = \frac{y-1}{4} = \frac{z-2}{4}$ is rotated through an angle $\frac{\pi}{2}$ about this line to contain the point (4, 3, 7) then which of the following statements is/are true ?
- (A) The equation of plane before rotation is $2x + 4y - 5z = 0$
- (B) The equation of plane after rotation is $2x - y - 5 = 0$
- (C) The equation of plane before rotation is $x - 2y + 2z - 1 = 0$
- (D) The equation of plane after rotation is $x + 2y + 2z = 1$
5. For the equation $|x^2 - a| = \cos^{-1}(\cos x)$ to have some solution the value of 'a' can be
- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $-\frac{1}{2}$ (D) $-\frac{1}{4}$
6. For ΔABC , let $A_1B_1C_1$ be the triangle formed by the excentres of ΔABC , let $\Delta A_2B_2C_2$ be formed by the excentres of $\Delta A_1B_1C_1$ and so on. Finally let $\Delta A_nB_nC_n$ be the n^{th} derived triangle that it is formed by the excentres of $\Delta A_{n-1}B_{n-1}C_{n-1}$ then
- (A) $\angle A_1 = 90^\circ - \frac{A}{2}$
- (B) The circumcentre of ΔABC always coincides with the orthocentre of $\Delta A_1B_1C_1$
- (C) as $n \rightarrow \infty$, $\Delta A_nB_nC_n$ tends to become equilateral
- (D) $B_1C_1 = a \cot \frac{A}{2}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Let O, O¹ and G be the circumcentre, orthocentre and centroid of ΔABC respectively. Now match the values in list-I and list-II.

list-I		list-II	
(A)	If S is any point in plane of A,B,C then $\frac{ \vec{SA} + \vec{SB} + \vec{SC} }{ \vec{SG} } =$	(P)	4
(B)	$\frac{ \vec{OA} + \vec{OB} + \vec{OC} }{ \vec{OO'} } =$	(Q)	3
(C)	$\frac{ \vec{O'A} + \vec{O'B} + \vec{O'C} }{ \vec{OO'} } =$	(R)	2
(D)	$\frac{ \vec{AO'} + \vec{O'B} + \vec{O'C} }{ \vec{AP} } =$	(S)	1
		(T)	1/2

(Where $|\vec{AP}|$ is diameter of circumcircle of ΔABC)

- (A) $A \rightarrow Q; B \rightarrow S; C \rightarrow R; D \rightarrow S$
- (B) $A \rightarrow Q; B \rightarrow R; C \rightarrow S; D \rightarrow S$
- (C) $A \rightarrow S; B \rightarrow S; C \rightarrow Q; D \rightarrow R$
- (D) $A \rightarrow S; B \rightarrow Q; C \rightarrow R; D \rightarrow R$

8.

List-I		List-II	
(A)	If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$, $\vec{c} = \vec{i} + 2\vec{j} - \vec{k}$ then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ is equal to	(P)	$\sqrt{3}$
(B)	If points P, Q and R have position vectors $\vec{r}_1 = 3\vec{i} - 2\vec{j} - \vec{k}$, $\vec{r}_2 = \vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{r}_3 = 2\vec{i} + \vec{j} - 2\vec{k}$ respectively relative to an origin O then the distance of P from the plane OQR is	(Q)	28
(C)	A line with direction ratios (2, 1, 2) intersects the lines $\vec{r} = -\vec{j} + \lambda(\vec{i} + \vec{j} + \vec{k})$ and $\vec{r} = -\vec{i} + \mu(2\vec{i} + \vec{j} + \vec{k})$ at A and B then AB is equal to	(R)	16
(D)	In a regular tetrahedron, the centres of four faces are the vertices of a smaller tetrahedron, The ratio of the volume of the smaller tetrahedron to that of the larger is $\frac{m}{n}$ where m and n are relatively prime positive integers. The value of m + n is	(S)	3

(A) $A \rightarrow R; B \rightarrow Q; C \rightarrow S; D \rightarrow Q$

(B) $A \rightarrow R; B \rightarrow S; C \rightarrow S; D \rightarrow Q$

(C) $A \rightarrow S; B \rightarrow Q; C \rightarrow S; D \rightarrow Q$

(D) $A \rightarrow Q; B \rightarrow R; C \rightarrow S; D \rightarrow Q$

9. Let $f(x) = \min\{\sec^2 \pi x, 3 - \sqrt{2} \sin \pi x\}$, $x \in (0, \frac{1}{2})$

List-I		List-II	
(A)	Number of integers in the range of $f(x)$ is	(P)	0
(B)	Number of points of local extremum of $f(x)$ is	(Q)	2
(C)	Number of values of x where $f(x)$ is an integer is	(R)	1
(D)	Number of solutions of the equation $f(x) = 1$ is	(S)	3
		(T)	4

(A) $A \rightarrow S; B \rightarrow R; C \rightarrow Q; D \rightarrow P$

(B) $A \rightarrow R; B \rightarrow Q; C \rightarrow Q; D \rightarrow S$

(C) $A \rightarrow R; B \rightarrow R; C \rightarrow R; D \rightarrow R$

(D) $A \rightarrow P; B \rightarrow S; C \rightarrow Q; D \rightarrow R$

10. Match the following

List-I		List-II	
(A)	$\sqrt{\sin x} : \left[0, \frac{\pi}{3}\right] \rightarrow [0, 1]$	(P)	one-one
(B)	$\frac{x+3}{x-1} : (1, \infty) \rightarrow (1, \infty)$	(Q)	many-one
(C)	$\sin x : \left[-\frac{\pi}{2}, \frac{4\pi}{3}\right] \rightarrow [-1, 1]$	(R)	into
(D)	$\frac{x^2}{x-2} : (2, \infty) \rightarrow [8, \infty)$	(S)	onto

(A) $A \rightarrow P, S; B \rightarrow P, R; C \rightarrow Q, S; D \rightarrow Q, S$

(B) $A \rightarrow Q, R; B \rightarrow P, S; C \rightarrow P, S; D \rightarrow Q, S$

(C) $A \rightarrow P, R; B \rightarrow P, S; C \rightarrow Q, S; D \rightarrow Q, S$

(D) $A \rightarrow P, R; B \rightarrow P, Q; C \rightarrow Q, S; D \rightarrow Q, S$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If Z is a non-real complex number such that $\arg(z^{1/3}) = \frac{1}{2} \arg(z^2 + \bar{z}z^{1/3})$ then the value of $|z|$ is
2. A circle concentric to the ellipse $\frac{4x^2}{289} + \frac{y^2}{\lambda^2} = 1$ ($\lambda < \frac{17}{2}$) passes through foci S_1, S_2 and cuts ellipse at point P . If area of ΔPS_1S_2 is 30 sq.units then $S_1S_2 =$
3. Let a, b, c are roots of $x^3 + x^2 - 2x + 1 = 0$ then the value of $\begin{vmatrix} a+c & b & a+2b+2c \\ a+b & c & b+2a+2c \\ b+c & a & c+2a+2b \end{vmatrix}$ is equal to
4. If $a_1 = 1$ and $a_n = n(1 + a_{n-1}) \forall n \geq 2$ then $\left(\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1} \right) \left(1 + \frac{1}{a_2} \right) \dots \left(1 + \frac{1}{a_n} \right) \right) \cdot \frac{1}{e} =$
5. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \leq 0$ and $f(x) - f(y) \leq 0$. The area of S is $k\pi$ then $k =$
6. Let $p, q^2 \in \mathbb{Q}$, the set of rational numbers and $\cos^2 \frac{\pi}{8}$ be a root of the equation $x^2 + px + q^2 = 0$ then the value of $[\cos p] + [\sin p] + p + 2 |\log_2 q|$ equals (Where $[\cdot]$ is G.I.F.)
7. Let $x, y \in \mathbb{R}$ such that $\cos x \cos y + 2 \sin y + 2 \sin x \cos y = 3$ then the value of $\tan^2 x + 5 \tan^2 y$ is
8. If $\lim_{n \rightarrow \infty} \left(\frac{3^n c_n}{2^n c_n} \right)^{\frac{1}{n}} = \frac{a}{b}$ where a, b are relatively prime positive integers then $\frac{a+b+1}{a-b} =$

SPACE FOR ROUGH WORK

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

SPACE FOR ROUGH WORK

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